Infinitely Many Prime Gaps Above Every Fixed Threshold

A classical factorial construction for OEIS A001223

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November 12, 2026

Abstract

We give a self-contained elementary argument that for every positive integer L, infinitely many consecutive-prime gaps have length at least $L+1$. The construction uses arbitrarily large blocks of consecutive composite integers. This is a classical consequence of Euclid's argument and does not resolve any conjecture about bounded prime gaps, their frequency, or specific even gap values.

Definitions and statement

Let $p_1 = 2 < p_2 = 3 < \cdots$ be the increasing primes, and let $g_n = p_{n+1} - p_n$. This is OEIS A001223, with $g_1 = 1$.

Theorem 1. *For every integer $L \geq 1$, there are infinitely many indices n with $g_n \geq L+1$. Consequently* $\limsup_{n\to\infty} g_n = +\infty$.

Proof. Fix $L \geq 1$ and put $M = (L+1)!$. For each positive integer t, the L consecutive integers

$$tM+2,\ tM+3,\ \ldots,\ tM+L+1$$

are composite: for each $2 \leq j \leq L+1$, the integer j divides M and hence divides $tM+j$, while $tM+j > j$.

Let p be the largest prime less than $tM+2$ and let q be the smallest prime greater than $tM+L+1$. Both exist: $2 < tM+2$, and there are infinitely many primes by Euclid's theorem. There is no prime in the displayed block; by the extremal choices there is no prime strictly between p and q. Thus p, q are consecutive primes and

$$q - p \geq (tM+L+2) - (tM+1) = L+1.$$

As $t \to \infty$, the lower bound $q > tM+L+1$ forces these consecutive-prime pairs to be infinitely many distinct pairs. Hence infinitely many indices n satisfy $g_n \geq L+1$. □

Scope and priority

This is the familiar factorial construction of arbitrarily long strings of composite numbers, and is included only as a rigorously documented property of A001223. It is not new, does not imply that a particular even gap repeats infinitely often, and does not address bounded gaps between primes (which concerns small gaps, not large ones).

References and verification scope

1. OEIS Foundation, *A001223: Prime gaps: differences between consecutive primes*, `https://oeis.org/A001223` (consulted November 12, 2026).

2. Euclid's infinitude of primes and the classical factorial composite-block construction.

AI disclosure. This document was generated and internally checked by ChatGPT (OpenAI), without independent human peer review. The argument is elementary and self-contained apart from Euclid's theorem.